

2. + parton shower (ISR + FSR)

standalone shower: event_PS.lhe

40 particles drawn (marker size $\sim \log p_T$; label = p_T [GeV] for $p_T > 5$ GeV)

b quark (status 52): $p_T = 78.12$ GeV, $\eta = -1.38$, $\phi = -2.73$

$\bar{b}$ quark (status 44): $p_T = 73.18$ GeV, $\eta = -3.41$, $\phi = 0.58$

net $b + \bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})| = 13.31$ GeV ($p_x = -10.21$, $p_y = 8.53$), $\Sigma|p_T| = 151.30$ GeV, $\Delta\phi(b, \bar{b}) = 2.98$

ISR (24 partons): $|\Sigma\vec{p}_T| = 64.39$ GeV ($p_x = 32.64$, $p_y = -55.51$)

FSR (14 partons (incl. a second $b\bar{b}$ pair from $g \rightarrow b\bar{b}$)): $|\Sigma\vec{p}_T| = 52.06$ GeV ($p_x = -22.43$, $p_y = 46.97$)

all 40 final-state partons: $|\Sigma\vec{p}_T| = 0.00$ GeV (the incoming partons carry no p_T , so the sum must vanish)

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